Composition: Q proves the uniqueness universal for the composed representation

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-incompleteness-representability-in-q/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author's 'using just a little bit of logic' phrase is retained; the logical step is the quantifier composition displayed in-span

Editorial note. The label prop:rep2 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Representability hypotheses for f and g. A_g represents g (both existence and uniqueness universals) and A_f represents f likewise.. Both uniqueness universals are the raw material; the proof is a logical composition of the two.

Proposition 0.1. *If* $h(n) = m$*, then* $\mathrm{Th}Q \vdash \forall z\,(A_h(\overline{n}, z) \to z = \overline{m})$.

Proof. Suppose $h(n) = m$, i.e., $f(g(n)) = m$. Let $k = g(n)$. Then

$$\mathrm{Th}Q \vdash \forall y\,(A_g(\overline{n}, y) \to y = \overline{k})$$

since A_g represents g, and

$$\mathrm{Th}Q \vdash \forall z\,(A_f(\overline{k}, z) \to z = \overline{m})$$

since A_f represents f. Using just a little bit of logic, we can show that also

$$\mathrm{Th}Q \vdash \forall z\,(\exists y\,(A_g(\overline{n}, y) \wedge A_f(y, z)) \to z = \overline{m}).$$

i.e., $\mathrm{Th}Q \vdash \forall y\,(A_h(\overline{n}, y) \to y = \overline{m})$. □